FLEET TEAM DIGEST

737MAX-FTD-29-18001

Issue Title : Hydraulic Pressure Module Cartridge Check Valve Fractures

Airplane Model	ATA	Minor Model(s)
737MAX	2911-71	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
737, 727, 737NG	9E991

Originated Date	Last Revised Date	Created On
05/03/2018	03/15/2019	03/12/2019

Estimated Completion Date	Next Update Date

Revision Description

- Confirmed release of CMM updates.

- Changed Issue Status to "Closed".

Applicability

No 737 MAX aircraft are delivered with the Parker H61C0552M1 valve, instead they are delivered with the Crissair P/N 1C4196 valve. However, there was approved 2-way interchangeability between the Crissair P/N 1C4196 valves and the Parker H61C0552M1 valves therefore interchange of these parts may have taken place in-service, hence the inclusion of the 737 MAX for this issue.

Description

This FTD discusses fractures of Parker P/N H61C0552M1 hydraulic pressure module cartridge check valves which may result in injury to personnel when hydraulic systems are pressurized.

In November 2017, Boeing received the first report of a P/N H61C0552M1 fracture with Serial Number (S/N) post S/N 21295. The fractured valve has under 50,000 flight cycles (FC). Boeing designated this a Personal Safety issue and initiated a Service Related Problem (SRP) investigation to address this concern. The investigation concluded that this check valve failure was due to insufficient strength of the aluminum check valve body which made it susceptible to impulse pressure loading that, in turn, resulted in fatigue failure.

The P/N H61C0552M1 Parker valves have been installed in Production on the System A pressure module in model 737-100/-200 airplanes, and on the System A and System B pressure modules in model 737-3/4/500 and 737NG airplanes until 2003. See illustration for reference. The optional Crissair P/N 1C4196 valve replaced the Parker valve in Production airplanes beginning in April 2003 but interchange of these parts may have taken place thereafter in-service.

Background

From 1992-1998, Boeing received a number of reports of check valve fractures. The majority of the fractures occurred in the first external thread of the valve housing nearest the hex head. Other operators reported valve fractures at the end flange and at the O-ring groove area. The subject is discussed in 737-200 ISAR 96-04-2906-30 "HYDRAULIC SYSTEM PRESSURE MODULE CHECK VALVE FRACTURE", 9 May 1996. Similar ISARs were published for 737-3/4/500 (96-06-2906-30) and 727 (96-03-2906-30). The same information was released in 737-FTD-29-05001.

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The valve supplier, Parker Hannifin, made a number of changes to the valve body. Fractures at the upper external threads were addressed by incorporating a larger radius thread relief. End flange fractures were addressed by increasing flange thickness and also the corner radius at the base of the end flange. The changes were accomplished within the existing Boeing specification design requirement envelope by tightening certain tolerances, and so the part numbers of the check valve and body were not changed. These changes were initially discussed in Parker Service Letter (SL) #6002, dated 09 March 1998.

Parker subsequently revised their SL in 2009 to advise operators that P/N 2671716 aluminum alloy valve bodies, made to Drawing Revision (Rev) R requirements and introduced at valve S/N 21295, have not had fractures since they were introduced, approximately ten years earlier. The revised Parker SL number is H61C0552M1-6002 Rev 3.

From 2009-2013, Boeing received nine (9) reports of body fractures in Parker P/N H61C0552M1 cartridge check valves which were installed in P/N 65-17821-xx hydraulic system pressure modules in 737CL airplanes. The fractured valves had accumulated between 25,000-44,000 FC and 36,0000-53,000 flight hours (FH). Available data indicates all were pre-S/N 21295 valves. In three (3) cases, the fractured head of the check valve damaged the wheel well structure at station 727 aft bulkhead structure. See attached photos and referenced Fleet Team Exchange (FTE) item 04957-737 for details.

In April 2010, Boeing released Rev C to SL #737-SL-29-108 on this subject to inform operators of the potential personal hazard associated with the Parker check valve fractures and to recommend that operators replace or rework Parker P/N H61C0552M1 check valves with serial numbers prior to 21295 in accordance with instructions in Parker SL H61C0552M1-6002 Rev 3. The Boeing SL also discussed optional use of Crissair P/N 1C4196 check valves with 15-5 PH CRES bodies which have been delivered in 737NG airplanes since 2003. There have been no reports of fractures in the Crissair check valves.

In January 2013, Boeing released 737-SL-29-108 Rev D which discussed the most recent reports of fractures in the Parker PN H61C0552M1 cartridge check valves. Boeing also designated these fractures as a Personal Safety issue and initiated an SRP investigation to address the issue. Special Attention Service Bulletins (SB) 737-29-1115 (737CL), 737-29-1114 (737NG) and P8-29-0003 (P-8A) were released, which recommended that operators replace all P/N H61C0552M1 valves with either P/N H61C0552M1 valves with S/N 21295 and higher which were life-limited to 50,000 FC or with the optional P/N 1C4196 valves which had no life limit restriction. Finally, 737-SL-29-108 Rev E was released in July 2013 to revise the SB numbers to note Rev 1 releases, the Boeing Action to discuss release of Rev 1 to the SBs, the Suggested Operator Action to correct typographical error in SB numbers, and the Crissair, Inc. contact information.

Status

In November 2017, Boeing received the first and only report of a P/N H61C0552M1 fracture post S/N 21295. The fractured valve had 36,716 FC. Boeing has designated this a Personal Safety issue and initiated an SRP investigation to address this concern. The failed part was not available for analysis thus making it impossible to definitively identify root cause of the failure. However, the failure mode (fracture), and the location of the fracture on the check valve body was similar to that of failed units that had, in the past, been evaluated by Boeing. Therefore, Boeing concluded that the November 2017 event was most likely a fatigue failure caused by impulse pressure loading.

In October 2018, Boeing released the new SB 737-29-1126 (737 MAX), as well as new SB 737-29-1127 (737CL), SB 737-29-1123 (737NG), and SB P8-29-0005 (P-8A) which replace SB 737-29-1115 (737CL), SB 737-29-1114 (737NG), and SB P8-29-0003 (P-8A), respectively. Moreover, Boeing issued SB 737-29-1114 R2, 737-29-1115 R2 and P8-29-0003 R2 advising operators to stop work on these, now obsolete, SBs. In addition, Boeing released Revision F to SL# 737-SL-29-0108 which discusses the November 2017 check valve failure and subsequent Boeing/Operator actions.

The new SBs direct operators to replace any Parker check valve P/N H61C0552M1 with a Crissair check valve, P/N 1C4196. The Crissair valves are manufactured with stainless steel which is not susceptible to fractures.

Boeing drawings have also been revised to show one way forward interchangeability with the Crissair check valve, P/N 1C4196.

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Furthermore, Boeing has released updated revisions of CMM 29-13-01 and CMM 29-13-03 to reflect the same one way forward interchangeability as the Boeing drawings.

Note: Boeing has released MOM-MOM-19-0122-01B to alert the 737 fleet of a Notice of Proposed Rulemaking (NPRM) 2018-NM-146-AD for this issue.

Interim Action

None

Final Action

- Release of new SB 737-29-1126 (737 MAX), SB 737-29-1127 (737CL), SB 737-29-1123 (737NG), and SB P8-29-0005 (P-8A)

- Release of 737-SL-29-108 Revision F

- Boeing drawing revision to show one way forward interchangeability with the Crissair check valve, P/N 1C4196

- CMM and IPC updates to reflect the same one way forward interchangeability as the revised drawings.

Milestones

Root Cause Established: March 2018 (COMPLETE)

Solution Selected: March 2018 (COMPLETE)

Service Bulletin Available: October 2018 (COMPLETE)

Parts Available: Ongoing

IPC: October 2018 (COMPLETE)

CMM: March 2019 (COMPLETE)

Operator Action

Boeing recommends that operators replace Parker P/N H61C0552M1 check valves in accordance with the instructions provided in the Boeing SB 737-29-1127 (737CL), SB 737-29-1123 (737NG), SB P8-29-0005 (P-8A), and SB 737-29-1126 (737 MAX). The new SB recommendations supersede recommendations previously given in SB 737-29-1114, SB 737-29-1115 and SB P8-29-0003. The recommended compliance time for replacement of Parker P/N H61C0552M1 check valves with the Crissair P/N 1C4196 check valves is 24 months after release of the new SBs.

Note: Reference 737 MAX AMM 29-11-71 for the appropriate procedures to remove/install check valves on-wing. See also Boeing CMMs 29-13-01 and 29-13-03 for off-airplane check valve installation into pressure modules.

Part Information

The P/N 1C4196 Crissair check valve replaces and is now one-way forward interchangeable with the P/N H61C0552M1 Parker check valve.

Affected Documents

Revision F to SL# 737-SL-29-0108, which discusses the most recent check valve failure and subsequent Boeing/Operator actions, has been released.

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737NG Master IPC 29-11-71 has been updated to reflect one way forward interchangeability with the P/N 1C4196 check valve. Similar updates for CMMs 29-13-01 and 29-13-03 have also been published.

Supplier Information

Supplier contact information for Parker check valve H61C0552M1 is as follows:

Parker Hannifin Corporation

Parker Aerospace

16666 Von Karman Avenue

Irvine, CA 92606-4917

Phone: 949-833-3000

FAX: 949-809-8390

E-mail: aerospacerfq@parker.com

Crissair check valve P/N 1C4196 is now being distributed through Aviall and can be purchased using the following contact information:

North America Sales and Customer Service

Phone: 1-800-AVIALL-1

FAX: 972-586-1361

International Sales and Customer Service

Phone: +1 972-586-1985

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Service Related Problem (SRP)	737-SRP-29-0130		02/08/2018
Service Related Problem (SRP)	737NG-SRP-29-0130		02/08/2018
Fleet Team Safety (FTS)	ISS-29-18-21806		
Component Maintenance Manual (CMM)	29-13-01		
Component Maintenance Manual (CMM)	29-10-38		
In Service Activity Report (ISAR)	96-04-2906-30 (737-1/200 airplanes)		05/09/1996

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Fleet Team Digest Article (FTD)	737-FTD-29-05001		
Fleet Team Economic (FTE)	737-FTE-29-X4957		03/27/2009
Service Letter (SL)	Parker 6002		03/09/1998
In Service Activity Report (ISAR)	96-06-2906-30 (737-3/4/500 airplanes)		04/08/1996
In Service Activity Report (ISAR)	96-03-2906-30 (727 airplanes)		03/29/1996
Service Letter (SL)	737-SL-29-108	F	10/17/2018
Service Letter (SL)	Parker H61C0552M1-6002	3	10/20/2009
Service Related Problem (SRP)	737-SRP-29-0084		10/18/2010
Fleet Team Safety (FTS)	FTS-29-11002		07/11/2012
Service Bulletin (SB)	737-29-1115	2	10/15/2018
Service Bulletin (SB)	737-29-1114	2	10/11/2018
Service Bulletin (SB)	P8-29-0003	2	10/19/2018
Component Maintenance Manual (CMM)	29-13-03		
Service Bulletin (SB)	SB 737-29-1127		10/08/2018
Service Bulletin (SB)	SB 737-29-1123		10/02/2018
Service Bulletin (SB)	SB 737-29-1126		10/02/2018
Service Bulletin (SB)	SB P8-29-0005		10/17/2018

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Check valve, cartridge	10-60491-3	Parker Hannifin	H61C0552M1
Pressure module	65-17821-xx		
Check valve body			2671716
Check valve, cartridge	S271A002-1	Crissair, Inc.	1C4196

Related Categories

- Fleet Team IdeaXchange : Topic of discussion on the Fleet Team IdeaXchange bulletin board
- SRPs - Safety : Service Related Problems in work by Boeing that have an airplane or personal safety impact

Attachments:

a) 727 Bulkhead Punctures

b) Cartridge Check Valve Fractures

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